# Medical Tubing

| Product / Application | BIS | ISO | ASTM | DIN | Other Standards |
|---|---|---|---|---|---|
| PVC Medical Tubing | IS 11497:2014; IS 12247; IS 13360 | ISO 10993 series (Biological evaluation of medical devices); ISO 3826 (Plastic collapsible containers for human blood and blood components, where applicable),ISO 16396 | ASTM D1599 (Short-Term Hydraulic Pressure Strength of Plastic Pipe, Tubing); ASTM D570 (Water Absorption of Plastics); ASTM D1785 (PVC Pipe/Tubing where applicable),ASTM D2122; ASTM D2240; ASTM F1386 | DIN EN ISO 10993 series (Biological evaluation of medical devices); DIN EN ISO 80369 series where connectors are applicable | - |
| Silicone Medical Tubing | IS 11497:2014; IS 12247; IS 13360 | ISO 10993 series; ISO 80369 series (small bore connectors where applicable),ISO 16396 | ASTM D412 (Rubber Tensile Properties); ASTM D624 (Tear Strength of Rubber); ASTM F2048 (Silicone Elastomer Evaluation where applicable),ASTM D1599; ASTM D2122; ASTM D2240; ASTM F1386 | DIN EN ISO 10993 series; DIN EN ISO 80369 series | - |
| Latex Medical Tubing | IS 11497:2014; IS 12247; IS 13360 | ISO 10993 series; ISO 10282 (if glove-grade latex application),ISO 16396 | ASTM D412; ASTM D624,ASTM D1599; ASTM D2122; ASTM D2240; ASTM F1386 | DIN EN ISO 10993 series | - |
| Polyurethane Medical Tubing | IS 11497:2014; IS 12247; IS 13360 | ISO 10993 series,ISO 16396 | ASTM D412; ASTM D624,ASTM D1599; ASTM D2122; ASTM D2240; ASTM F1386 | DIN EN ISO 10993 series | - |
| Oxygen Tubing | IS 11497:2014; IS 12247; IS 13360 | ISO 5367 (Breathing tubes and connectors),ISO 10993; ISO 16396 | ASTM D412; ASTM D1599,ASTM D2122; ASTM D2240; ASTM F1386 | DIN EN ISO 5367 (Breathing tubes and connectors) | IEC 60601-1 where supplied as medical electrical equipment accessory |
| Suction Tubing | IS 11497:2014; IS 12247; IS 13360 | ISO 8836 (Suction catheters and suction equipment); ISO 10993 series,ISO 16396 | ASTM D412; ASTM D1599,ASTM D2122; ASTM D2240; ASTM F1386 | DIN EN ISO 8836 (Suction catheters and equipment) | - |
| Dialysis Tubing | IS 11497:2014; IS 12247; IS 13360 | ISO 8637 series (Extracorporeal blood and fluid circuits); ISO 10993 series,ISO 16396 | ASTM D412; ASTM F88 (Packaging); ASTM F1980,ASTM D1599; ASTM D2122; ASTM D2240; ASTM F1386 | DIN EN ISO 8637 series (Extracorporeal blood and fluid circuits) | AAMI RD standards (Renal dialysis applications where applicable) |
| Infusion Pump Tubing | IS 11497:2014; IS 9824 | ISO 8536 series (Infusion equipment); ISO 10993 series,ISO 1135 | ASTM D412; ASTM D1599,ASTM D4169; ASTM F2503 | DIN EN ISO 8536 series (Infusion equipment for medical use) | IEC 60601-2-24 |
| Enteral Feeding Tube | IS 11497:2014; IS 12247; IS 13360 | ISO 80369-3 (Enteral connectors); ISO 10993 series,ISO 16396 | ASTM F640 (Radiopacity); ASTM D412,ASTM D1599; ASTM D2122; ASTM D2240; ASTM F1386 | DIN EN ISO 80369-3 (Enteral connectors); DIN EN ISO 10993 series | AAMI/ANSI CN27 |
| Nasogastric Tube | IS 11497:2014; IS 12247; IS 13360 | ISO 20696 (Urinary catheters, applicable principles); ISO 10993 series,ISO 16396 | ASTM F640; ASTM D412,ASTM D1599; ASTM D2122; ASTM D2240; ASTM F1386 | DIN EN ISO 10993 series; DIN EN ISO 80369-3 where applicable | - |
| Tracheostomy Tube | IS 15998,IS 12247; IS 13360 | ISO 5366 (Tracheostomy tubes),ISO 10993; ISO 16396 | ASTM F640; ASTM D412,ASTM D1599; ASTM D2122; ASTM D2240; ASTM F1386 | DIN EN ISO 5366 (Tracheostomy tubes) | - |
| Endotracheal Tube | IS 15998,IS 12247; IS 13360 | ISO 5361 (Tracheal tubes and connectors),ISO 10993; ISO 16396 | ASTM F640; ASTM D412,ASTM D1599; ASTM D2122; ASTM D2240; ASTM F1386 | DIN EN ISO 5361 (Tracheal tubes and connectors) | AAMI/ANSI ISO 18190 adoption (Airway devices where applicable) |
| Chest Drain Tube | IS 12247; IS 13360 | ISO 7197 (Neurosurgical drainage catheters, applicable principles); ISO 10993 series,ISO 16396 | ASTM F640; ASTM D412,ASTM D1599; ASTM D2122; ASTM D2240; ASTM F1386 | DIN EN ISO 10993 series; DIN EN ISO 80369 where applicable | - |
| Oxygen Cannula | IS 11497:2014; IS 12247; IS 13360 | ISO 13544 (Oxygen therapy equipment, where applicable); ISO 10993 series,ISO 16396 | ASTM D412; ASTM F640,ASTM D1599; ASTM D2122; ASTM D2240; ASTM F1386 | DIN EN ISO 10993 series | - |
| Extension Tubing | IS 11497:2014; IS 12247; IS 13360 | ISO 80369 series; ISO 8536 series,ISO 10993; ISO 16396 | ASTM D1599; ASTM D412,ASTM D2122; ASTM D2240; ASTM F1386 | DIN EN ISO 8536 series; DIN EN ISO 80369 series | - |

*Source field: Medical devices*